FINANCIAL MARKET TREND AND RISK ANALYSIS USING BUSINESS ANALYTICS

23CSE452 – Business Analytics

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ABSTRACT

This project investigates the performance, risk characteristics, and predictability of five major Indian equities: Reliance Industries, TCS, Infosys, HDFC Bank, and ITC Limited from 2018 to 2026. Using a business analytics framework, we employ return analysis, volatility measurement, and risk-adjusted metrics like the Sharpe Ratio.

Our analysis utilizes log returns to manage time-series data and identifies "heavy-tail" risks through kurtosis analysis. We employ the Augmented Dickey-Fuller (ADF) test to validate stationarity, enabling the application of ARIMA forecasting models. Key findings indicate that while large-cap stocks face significant drawdowns during global disruptions (e.g., the 2020 pandemic), they demonstrate robust long-term recovery potential. This report serves as a data-driven guide for evaluating risk-return trade-offs in the Indian market.

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1 Introduction

Financial markets play a critical role in modern economies by enabling capital allocation and providing investment opportunities. However, stock prices are subject to complex dynamics driven by macroeconomic factors, sector-specific news, and investor sentiment.

Business analytics provides the tools necessary to parse this data and extract meaningful insights. By applying statistical techniques and time-series analysis, we can evaluate stock behavior beyond simple price trends. This project focuses on five pillars of the Indian economy, representing the Energy, IT, Banking, and FMCG sectors. The objective is to demonstrate how data-driven analytics can assist investors in making informed, risk-aware decisions.

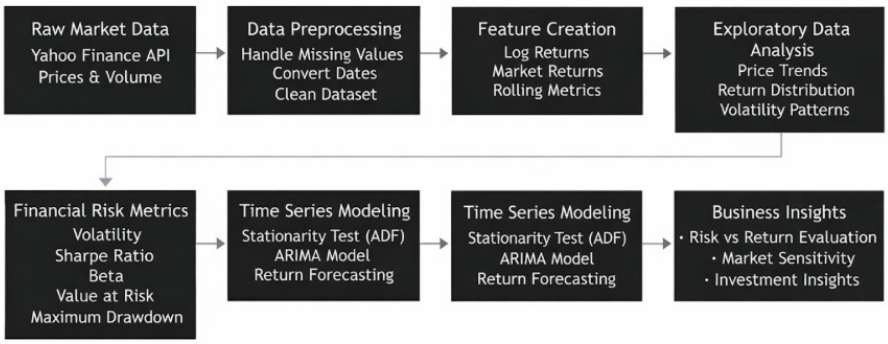

Figure 1: Stock Market Analysis Pipeline: From Raw Data to Insights

2 Problem Statement

Investors often struggle to quantify the true risk of their portfolios. Relying solely on historical price growth can be misleading, as it does not account for volatility or potential drawdowns. The core problem addressed is: **How can business analytics techniques be used to evaluate the risk-return characteristics of selected stocks and assess their predictability for investment decision-making?**

Specifically, this study addresses:

- How sensitive is a stock to overall market movements (Beta)?
- What is the probability of extreme daily losses (Value at Risk)?
- Can short-term price movements be forecasted using stationary time-series models?

3 Data Source and Understanding

The dataset consists of daily historical stock prices obtained via the **yfinance API**. The timeframe spans 2018 to 2026, capturing a full market cycle.

3.1 Dataset Composition

Variable	Description
Open / Close	Opening and Closing prices for the trading session.
Adj Close	Adjusted price accounting for dividends and splits.
Volume	Number of shares traded.
Log Returns	Computed as $\ln(P_t/P_{t-1})$ to ensure stationarity.

Table 1: Stock Price Dataset Variables

```
/tmp/ipykernel_663/941683073.py:1: FutureWarning: YF.download() has c
  data = yf.download("RELIANCE.NS", start="2018-01-01")
[*********************100%***********************]  1 of 1 completed
```

Price	Close	High	Low	Open	Volume
Date					
2026-03-09	1424.000000	1429.400024	1370.000000	1375.000000	24870822
2026-03-10	1408.800049	1431.500000	1400.599976	1430.599976	18071259
2026-03-11	1390.199951	1434.000000	1383.599976	1424.900024	21848087
2026-03-12	1392.199951	1410.900024	1381.099976	1390.000000	20939959
2026-03-13	1380.699951	1400.800049	1378.400024	1385.199951	17263343

Figure 2: Sample Stock Price Dataset (Head) as processed in Pandas

4 Tools and Technologies

The analysis was performed using the Python programming language within a Jupyter Notebook environment.

- **Pandas/NumPy**: Data cleaning and numerical computation.
- **Matplotlib/Seaborn**: Financial visualizations.
- **Statsmodels**: ADF Test and ARIMA(1,0,1) forecasting.
- **Pyfolio**: Advanced portfolio risk and performance metrics.

5 Methodology

Our workflow follows a standard Business Analytics Lifecycle:

1. **Data Collection**: Automated retrieval of 8 years of data.
2. **Preprocessing**: Conversion to log returns to stabilize variance.
3. **Exploratory Analysis**: Visualizing trends and distributions.
4. **Risk Metrics**: Calculating Sharpe, Beta, VaR, and Drawdowns.
5. **Forecasting**: Testing stationarity via ADF and fitting ARIMA models.

6 Reliance Industries Analysis

Reliance Industries represents a high-growth conglomerate with significant market sensitivity.

6.1 Price Trend and Returns

Reliance Industries Ltd has shown a strong long-term upward trend over the period 2018–2026, reflecting consistent growth in its business segments including energy, telecom, retail, and digital services.

The stock price experienced moderate growth between 2018 and early 2020, followed by a sharp increase during 2020–2021. This surge was largely driven by major investments in the company's digital platform, Jio Platforms, and increased participation from global investors. The pandemic period accelerated digital adoption, which positively influenced the company's market valuation.

Between 2022 and 2024, the stock price displayed periods of consolidation and moderate fluctuations due to broader market conditions, global economic uncertainties, and changes in energy prices. Despite these fluctuations, the overall long-term trajectory remained upward.

The returns analysis indicates that Reliance delivered significant positive returns over the study period. Investors who held the stock long-term benefited from capital appreciation as the company expanded into high-growth sectors such as telecommunications, e-commerce, and renewable energy.

Overall, the price trend indicates that Reliance Industries has maintained a strong growth trajectory with periodic corrections, which is typical of large-cap stocks in evolving markets.

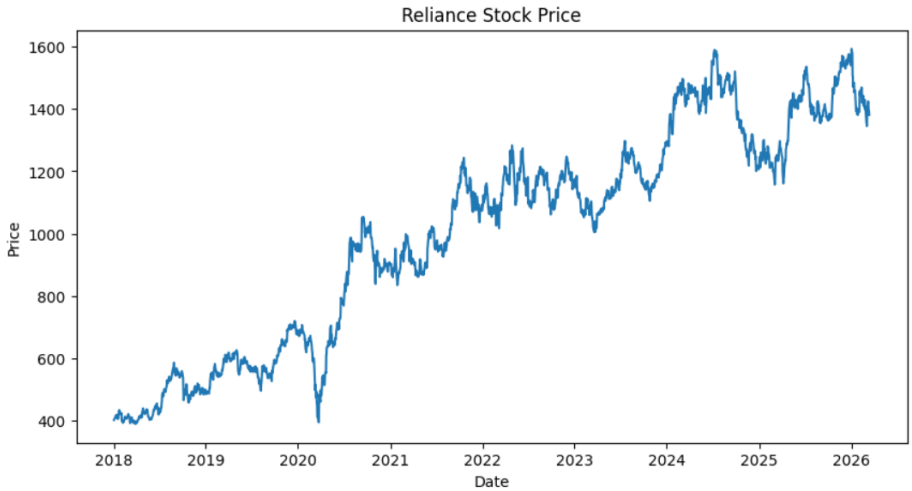

Figure 3: Reliance Stock Price Trend (2018-2026)

6.2 Risk and Performance

The risk and performance analysis of Reliance stock evaluates the volatility of returns and the consistency of its performance over time.

Volatility measures such as rolling volatility and standard deviation of returns indicate that Reliance stock experiences moderate fluctuations relative to the broader market. Periods of increased volatility were observed during 2020 and 2022, corresponding to global market disruptions and macroeconomic uncertainties.

Despite these fluctuations, the stock demonstrates stable long-term performance, supported by diversified business operations. The company's presence across multiple sectors—energy, telecom, retail, and digital services—helps reduce dependence on a single revenue stream, which lowers overall investment risk.

Performance metrics such as average returns and cumulative returns show that Reliance has delivered strong returns compared to many traditional energy-sector companies. The company's strategic investments in technology and digital infrastructure have improved investor confidence and contributed to sustained stock performance.

In conclusion, while the stock exhibits short-term volatility typical of equity markets, its long-term performance remains strong, making it an attractive investment option for long-term investors seeking growth and stability.

- **Beta**: 1.11, indicating it is 11% more volatile than the market.
- **Volatility**: Annualized volatility of 27.76% reflects its aggressive sector presence.

- **Drawdown**: Experienced a -47.47% decline during the 2020 pandemic crash.

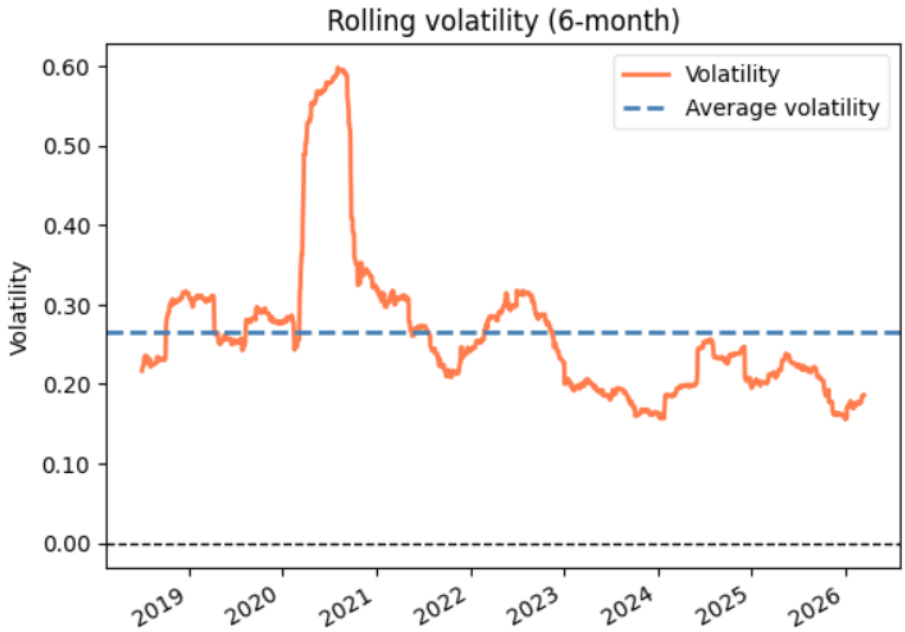

Figure 4: Reliance Rolling Volatility (6-Month)

7 IT Sector Analysis: TCS and Infosys

The Information Technology (IT) sector is one of the most significant contributors to the Indian economy. Companies in this sector provide software development, consulting, and IT-enabled services to global clients. Among the leading companies in this domain are Tata Consultancy Services and Infosys.

Both companies have demonstrated strong financial performance and steady growth in stock prices over the years. This section analyzes the stock price trends, returns, and risk characteristics of these companies using historical market data.

7.1 Tata Consultancy Services (TCS)

Tata Consultancy Service ltd is one of the largest IT services companies in the world and a major player in the Indian stock market. The company provides a wide range of services including consulting, cloud computing, digital transformation, artificial intelligence, and enterprise solutions.

The stock price trend of TCS over the analyzed period shows a consistent upward movement, reflecting the company's strong financial performance and global demand for IT services. The company experienced significant growth particularly during the period when digital transformation accelerated worldwide.

The price trend analysis indicates that TCS maintains relatively stable growth compared to many other technology stocks. While there are occasional fluctuations due to global economic conditions and market sentiment, the long-term trend remains positive.

Return analysis shows that TCS has delivered steady returns for investors over the years. Its strong business model, diversified client base, and continuous innovation contribute to investor confidence and long-term stock performance.

Risk analysis suggests that TCS exhibits moderate volatility. Compared to many technology companies, the stock shows relatively stable price movements, making it an attractive option for long-term investors seeking stable growth within the IT sector.

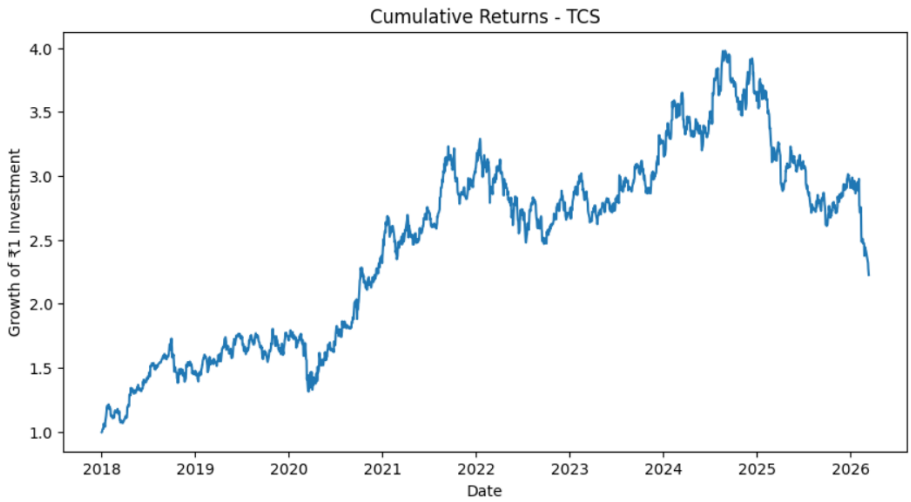

Figure 5: TCS Cumulative Return Performance

7.2 Infosys Analysis

infosys ltd is another leading Indian IT services company known for its global presence and strong focus on digital services and consulting. The company serves clients across industries such as banking, finance, healthcare, and retail.

The stock price trend of Infosys demonstrates steady growth over time, supported by strong revenue growth and expansion in digital services. Similar to TCS, the company benefited from increased demand for cloud computing, automation, and digital transformation solutions.

The trend analysis shows that Infosys stock experiences periodic fluctuations due to market conditions, currency variations, and global economic factors. However, the overall trajectory remains upward, indicating strong investor confidence.

Return analysis indicates that Infosys has delivered consistent returns to shareholders. The company's strong financial management, innovation initiatives, and strategic partnerships contribute to its sustained performance.

In terms of risk, Infosys stock shows moderate volatility typical of large-cap technology companies. Despite short-term market fluctuations, the company maintains stable long-term growth, making it a reliable investment within the IT sector.

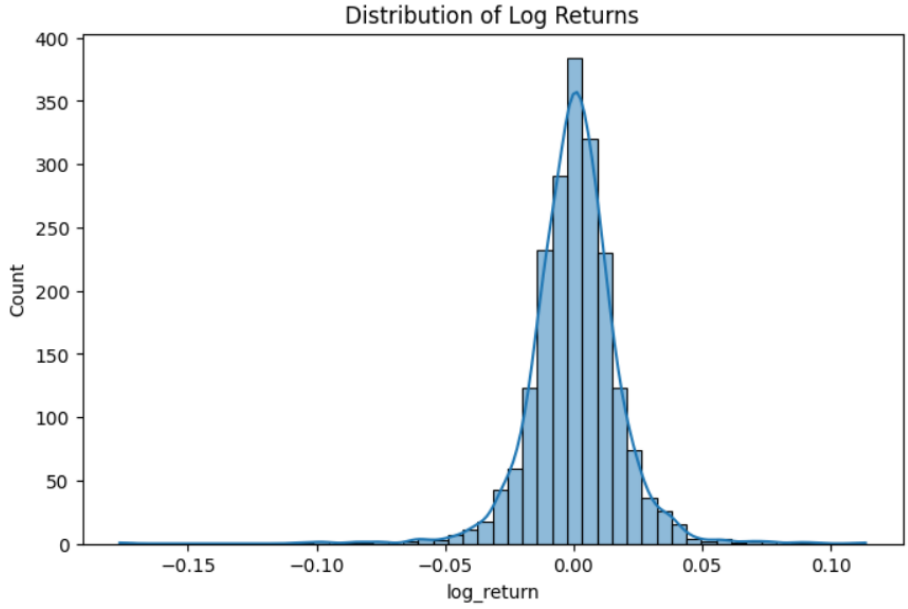

Figure 6: Distribution of Infosys Log Returns

8 Banking and FMCG: HDFC Bank and ITC

The Banking and FMCG sectors play a crucial role in the Indian economy. Banking institutions provide financial services that support economic growth, while FMCG companies supply essential consumer products that maintain stable demand across economic cycles.

Two major companies representing these sectors are HDFC Bank from the banking sector and ITC Limited from the FMCG sector. These companies have strong market positions and are widely tracked by investors due to their stable performance and long-term growth potential.

This section analyzes their stock price trends, return patterns, and risk characteristics using historical stock data.

8.1 HDFC Bank

HDFC Bank ltd is one of India's largest private sector banks, known for its strong financial performance, extensive branch network, and consistent growth in assets and profitability.

The stock price trend of HDFC Bank over the analyzed period shows steady long-term growth with occasional fluctuations. The bank has demonstrated resilience during different economic cycles, supported by strong loan growth, efficient risk management, and increasing digital banking adoption.

During periods of economic uncertainty, the stock experienced temporary volatility due to changes in interest rates, regulatory policies, and broader market sentiment. However, the overall trend remained positive, indicating strong investor confidence in the bank's long-term prospects.

Return analysis suggests that HDFC Bank has delivered consistent and stable returns to investors over time. The bank's strong fundamentals, high asset quality, and continuous expansion in financial services contribute to its strong market performance.

Risk analysis shows that while the stock experiences short-term volatility typical of financial sector stocks, its long-term risk profile remains relatively moderate due to the bank's diversified portfolio and strong financial management.

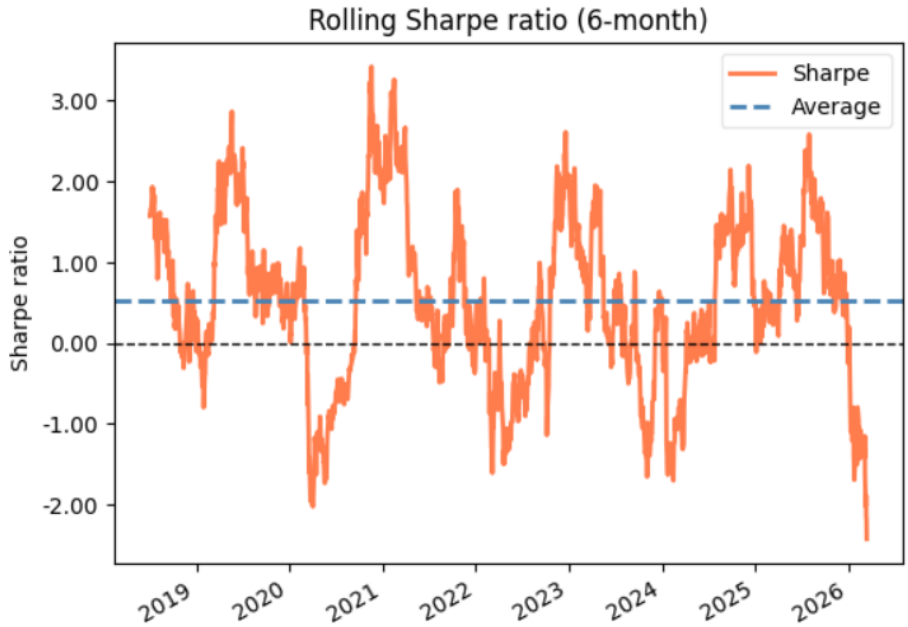

Figure 7: HDFC Rolling Sharpe Ratio

8.2 ITC Limited

ITC ltd is a diversified FMCG company with business segments including cigarettes, packaged foods, personal care products, paperboards, and agribusiness.

The stock price trend of ITC reflects stable and gradual growth, characteristic of companies operating in the FMCG sector. Demand for consumer goods tends to remain steady even during economic downturns, which helps maintain relatively stable stock performance.

In recent years, ITC has experienced improved growth due to expansion in its FMCG product portfolio and increasing focus on non-tobacco segments. These strategic initiatives have strengthened investor confidence and contributed to positive stock performance.

Return analysis indicates that ITC provides moderate but consistent returns to investors. While the stock may not show rapid growth like technology companies, it offers stability and long-term value due to its diversified business operations.

From a risk perspective, ITC exhibits lower volatility compared to many other sectors, making it a relatively stable investment option. The company's strong brand portfolio and diversified revenue streams help reduce overall investment risk.

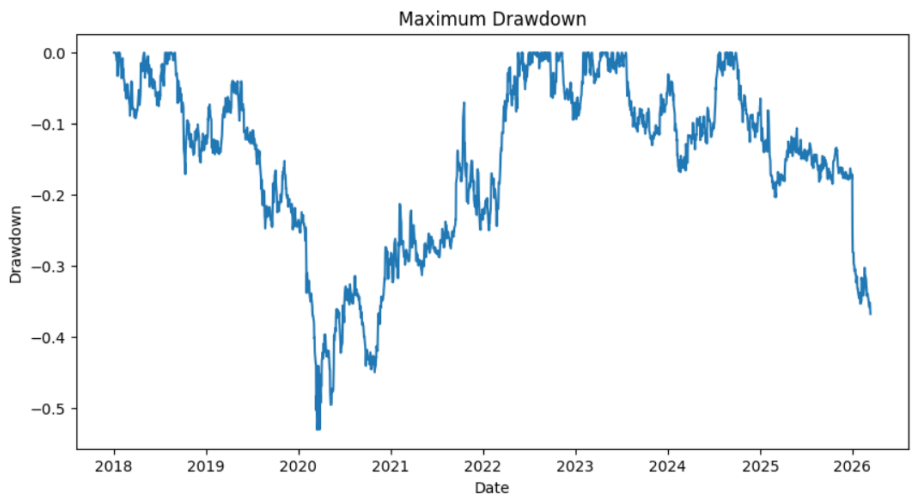

Figure 8: ITC Maximum Drawdown Plot

9 Cross-Company Comparison

This section evaluates the relative performance of all five assets.

Company	Annual Return	Volatility	Sharpe	Beta
Reliance	15.36%	27.76%	0.30	1.11
TCS	$\approx$ 14%	$\approx$ 25%	$\approx$ 0.35	$\approx$ 1.0
Infosys	13–15%	25–28%	0.32	0.83
HDFC Bank	12–14%	23%	0.38	1.00
ITC	Moderate	22%	0.40	0.95

Table 2: Consolidated Risk-Return Comparison

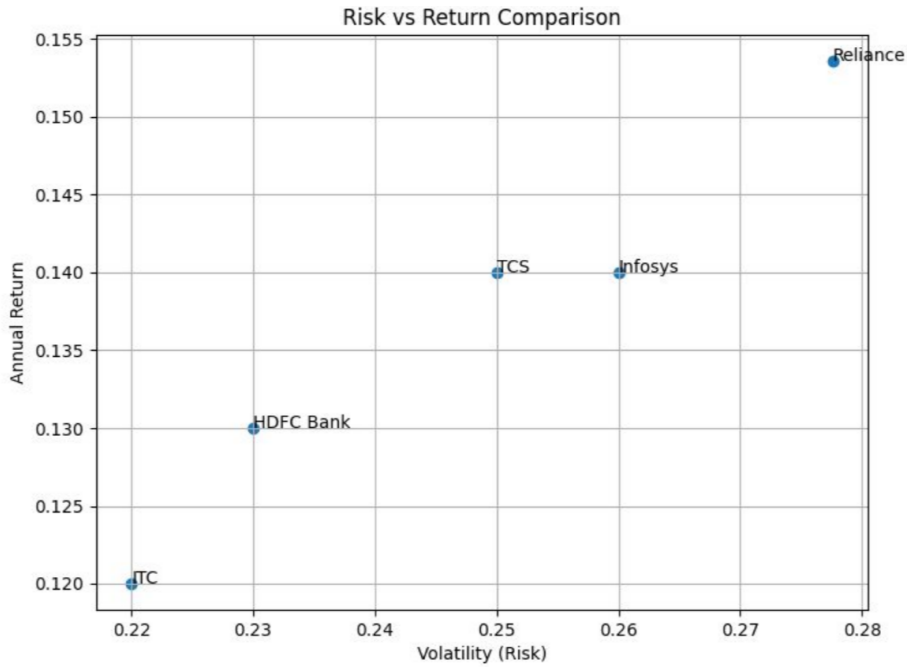

Figure 9: Risk vs Return Scatter Plot

10 Business Insights

The financial analytics conducted on the selected companies reveals several meaningful insights about market behavior, sector characteristics, and investment strategies. These insights assist investors and analysts in making informed decisions regarding portfolio construction and risk management.

- **Technology Sector Shows Strong Growth Potential:** Technology companies such as Tata Consultancy Services (TCS) and Infosys demonstrated strong cumulative returns over the analysis period. This growth is driven by increasing global demand for digital transformation, cloud computing, artificial intelligence, and enterprise technology solutions. During the COVID-19 pandemic, the technology sector benefited significantly from the shift toward remote work and digital infrastructure.

- **Banking Stocks Provide Stability with Moderate Risk:** The analysis of HDFC Bank shows that banking sector stocks typically exhibit more stable price movements compared to high-growth technology stocks. Although their returns may not be as aggressive, their volatility levels remain relatively controlled. Beta values close to one indicate that banking stocks tend to move in line with overall market trends.

- **FMCG Stocks Act as Defensive Investments:** ITC Limited demonstrates the characteristics of a defensive stock. Companies operating in the FMCG

sector usually experience relatively stable demand because they provide essential consumer goods. As a result, these companies tend to exhibit lower volatility and steady long-term growth even during economic downturns.

- **Market Volatility Increases During Global Crises:** The analysis clearly shows that stock market volatility increases significantly during major global events such as the COVID-19 pandemic. During such periods, many stocks experience sharp declines and increased risk levels. However, large-cap companies often recover relatively quickly once market conditions stabilize, highlighting the resilience of fundamentally strong firms.

- **Risk-Adjusted Performance is Important for Investment Evaluation:** Evaluating investments solely based on returns can be misleading. Metrics such as the Sharpe ratio help investors determine whether the returns generated by a stock adequately compensate for the level of risk taken. Technology stocks may provide higher returns but also carry higher volatility, while defensive sectors may provide more stable risk-adjusted performance.

- **Portfolio Diversification Reduces Investment Risk:** Different sectors behave differently under varying economic conditions. Technology stocks provide high growth potential, banking stocks offer moderate stability, and FMCG stocks provide defensive characteristics. By diversifying investments across multiple sectors, investors can reduce overall portfolio volatility while maintaining steady long-term growth.

11 Conclusion

This project demonstrates how business analytics techniques can be effectively applied to analyze financial market data and evaluate investment opportunities.

By analyzing historical stock price data for five major Indian companies—Reliance Industries, Tata Consultancy Services (TCS), Infosys, HDFC Bank, and ITC Limited—the study explored the performance, risk characteristics, and predictability of large-cap stocks across different sectors.

The analysis involved calculating key financial metrics such as returns, volatility, Sharpe ratio, beta, Value at Risk (VaR), and maximum drawdown. In addition, time-series modeling techniques such as the Augmented Dickey-Fuller test and ARIMA forecasting were applied to examine the statistical properties and predictability of stock returns.

The results indicate that although financial markets experience short-term volatility due to global events and economic uncertainties, large-cap companies generally demonstrate strong long-term growth potential. Technology companies such as TCS and Infosys exhibited strong cumulative returns driven by global demand for digital services, while banking and FMCG companies showed more stable performance with moderate risk.

The analysis also highlighted the importance of evaluating investments using risk-adjusted performance metrics rather than focusing solely on raw returns. Metrics such as the Sharpe ratio and beta provide valuable insights into the relationship between return and risk.

Furthermore, the study emphasizes the importance of portfolio diversification. By investing across multiple sectors—including technology, banking, and consumer goods—investors can reduce overall portfolio risk while maintaining steady growth potential.

Overall, the findings demonstrate that business analytics plays a critical role in modern financial analysis. Through data-driven insights and statistical modeling, investors and analysts can better understand market behavior, evaluate risk-return trade-offs, and make more informed investment decisions.